The Effect of Socioeconomic and Food Access Factors on Chronic Health Outcomes in Lancaster County, NE

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Abstract

This study looks at the relationship between food accessibility, socioeconomic factors, and chronic health outcomes in Lancaster County, Nebraska. We focus on obesity, type II diabetes, and coronary heart disease. Traditional food desert research treats physical distance to grocery stores as the main driver of poor health. Our analysis tells a more complex story. We combined spatial data from the USDA and OpenStreetMap with health and economic data from the CDC and the U.S. Census Bureau. Then we built a multiple linear regression model to measure these relationships at the census tract level.

We found that socioeconomic variables, particularly educational attainment, predict chronic health outcomes far better than physical proximity to food outlets. Percent without a high school diploma correlated with obesity at $r \approx 0.75$. A random forest model confirmed this, assigning educational attainment an importance weight of 0.721 compared to 0.143 for distance to the nearest grocery store. Distance to food was a consistently weak predictor across every model we tested. Spatial mapping shows geographic clusters of health risk that align with economic disadvantage rather than store density. Public health interventions should focus on reducing financial and educational barriers rather than placing new grocery stores.

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Chapter 1

Introduction

1.1 Motivation

Public health discussions often focus on food deserts and food swamps. These are areas where physical distance to food sources is thought to be a primary barrier to healthy eating. That simple geographic explanation misses the complex reality of how people live and access food. Lincoln, NE has a dense urban core alongside sprawling rural outskirts. The relationship between where you live and how healthy you are is rarely simple [3].

Lancaster County, NE scored 8.1 out of 10 on the *Food Environment Index* (FEI) for 2025, per the University of Wisconsin Population Health Institute's *County Health Rankings and Roadmaps* [2]. That index measures factors that contribute to a healthy food environment. The score ranks high compared to Nebraska (7.7) and the United States (7.6). But deeper indicators under Population Health and Well Being and Social and Economic Factors show a more concerning picture.

Food Insecurity: 15%
Physical Inactivity: 21%
Adult Obesity: 36%

That gap raises our main question. Do hidden drivers affect health outcomes like obesity, type II diabetes, and coronary heart disease? And why do they not show up in the FEI score?

1.1.1 Central Question and Problem

This project aims to find the strongest predictors of chronic health outcomes. We test whether physical distance to a store is truly the main problem or whether socioeconomic factors like income, education, and vehicle access drive health disparities more. By analyzing Lancaster County at the census tract level, we want to identify the structural barriers that keep residents from reliably accessing nutritious food.

1.1.2 Importance

Understanding these factors matters for local policymakers and health organizations. If obesity in Lincoln comes from proximity to food, building more stores is the solution. If socioeconomic conditions drive it, the solution might involve improving public transit or expanding economic support for specific communities. That distinction directly affects how resources get allocated and which efforts are likely to work [4, 3].

1.1.3 Real-World and Mathematical Significance

From a real world perspective, this project identifies vulnerability hotspots where health risks are highest. That lets policymakers move past the one size fits all food desert definition. From a mathematical perspective, we use multiple linear regression and random forest modeling to measure how much of the variation in health outcomes socioeconomic and geographic variables can explain. An $R^2 \approx 0.49$ in the linear model means nearly half of a census tract's health profile is predictable from a small set of socioeconomic indicators. That is a meaningful result in applied social science.

1.2 Overview of the Project

This study combines health data from the CDC with socioeconomic data from the U.S. Census Bureau and spatial data from OpenStreetMap. We start with a mathematical foundation in correlation and multiple linear regression, then use spatial mapping to visualize patterns across Lancaster County. Chapter 2 covers the data and modeling assumptions. Chapter 3 provides mathematical background. Chapter 4 describes the analysis. Chapter 5 presents our findings and conclusions.

1.2.1 Key Terms and Definitions

1. **Food Insecurity:** Limited or uncertain access to adequate food [3].

2. **Food Desert:** An area that lacks a grocery store, or where consumers must travel significant distances to reach one [12].

3. **Census Tract:** A small, relatively permanent subdivision of a county the U.S. Census Bureau uses for statistical data collection. Census tracts aim to be fairly uniform in population characteristics, economic status, and living conditions [13].

Chapter 2

Data and Assumptions

2.1 Data

This study combines several datasets to examine how socioeconomic conditions and food accessibility relate to chronic health outcomes across census tracts in Lancaster County, Nebraska. Each census tract is one unit of observation throughout the analysis.

We gathered socioeconomic data from the American Community Survey (ACS) through the U.S. Census Bureau [1]. The ACS variables we used are percent uninsured, percent of households without access to a vehicle, percent without a high school diploma, and poverty rate. These variables give tract level measures of economic vulnerability, educational disadvantage, and transportation limitations that may restrict food access and health outcomes. We pulled data from the following ACS tables.

1. **S1501** for Educational Attainment, which gave us percent without a high school diploma

2. **S2701** for Health Insurance Coverage, which gave us percent uninsured

3. **B08201** for Household Size by Vehicles Available, which gave us percent without a car

4. **S1701** for Poverty Status in Past 12 Months, which gave us poverty rate

We got health outcome data from the CDC PLACES data portal [8]. Our primary health variables were obesity prevalence, diabetes prevalence, and coronary heart disease prevalence. Here is how we define each one.

1. **Obesity:** A body mass index (BMI) greater than 30.0 kg/m^2 [7].

2. **Type II Diabetes:** A condition where cells cannot absorb blood sugar due to insulin resistance, preventing normal glucose uptake [9].

3. **Coronary Heart Disease:** Plaque buildup within the arteries that narrows them and can partially or fully restrict blood flow to the heart [6].

These three serve as our response variables. We chose all three because they share similar dietary and food access risk factors.

We merged all datasets at the census tract level using a common geographic identifier (GEOID). Preprocessing steps included standardizing tract identifier formats across sources, converting string-formatted percentages to numeric values, handling missing values, and building derived variables such as fast food density, grocery density, and distance-based access measures.

We built food access variables using geographic data from OpenStreetMap, processed with OSMnx [5]. This gave us locations of grocery stores, supermarkets, convenience stores, and fast food establishments in Lancaster County. We calculated distance to the nearest grocery store from each tract's centroid using straight-line distance in projected coordinates, then converted to kilometers. We calculated density variables by spatially joining food locations to census tracts, counting relevant locations within each tract, and dividing by tract area in square kilometers.

2.2 Assumptions and Limitations

1. We treat census tracts as independent observations. In practice, neighboring tracts may share similar characteristics, so this is a simplification. It lets us apply standard regression and correlation methods at the tract level.

2. The multiple linear regression model assumes an approximately linear relationship between obesity prevalence and the explanatory variables, with error terms of mean zero and roughly constant variance. Residual analysis suggests this is a reasonable first approximation.

3. We use geographic distance to grocery stores as a proxy for food access. True food access also depends on transportation availability, transit coverage, affordability, and individual mobility, so this measure is useful but incomplete.

4. We aggregate data at the census tract level. Tract level averages may hide variation within neighborhoods. Read our conclusions as tract level patterns rather than individual level claims.

5. We assume OpenStreetMap food location data reasonably represents the local food environment. Some store locations may be missing or inconsistently categorized. We used a broader set of grocery-related categories when building food access variables to reduce this limitation.

6. This study covers Lancaster County, Nebraska only. The relationships we found may not apply to counties with different urban structures, transportation systems, or socioeconomic compositions.

Chapter 3

Mathematical Background

This chapter introduces the mathematical concepts we used to analyze relationships between socioeconomic factors, food access, and health outcomes.

3.1 Correlation

We use the Pearson correlation coefficient to measure the strength of the linear relationship between two variables [10]. For two variables X and Y, the correlation is

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

The value of r lies in the interval $[-1, 1]$. When $r \approx 1$, there is a strong positive linear relationship. When $r \approx -1$, there is a strong negative relationship. When $r \approx 0$, there is little to no linear relationship. We used correlation analysis as an initial screening tool to find which variables associated most strongly with obesity before fitting regression models.

3.2 Multiple Linear Regression

We used a multiple linear regression model to relate obesity prevalence to several explanatory variables at once.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i,$$

where:

- Y_i is the obesity rate in tract i,
- X_{1i} represents percent without a high school diploma,
- X_{2i} represents distance to the nearest grocery store,
- X_{3i} represents fast food density,
- $\beta_0, \beta_1, \beta_2, \beta_3$ are model coefficients,
- ε_i is an error term assumed to be independent with mean zero and constant variance.

The model gives us coefficient estimates that describe the direction and relative size of each predictor's association with obesity. This provides an interpretable baseline for comparing socioeconomic and geographic factors.

3.3 Coefficient of Determination

The coefficient of determination, R^2, measures the proportion of variance in the response that the model explains.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

A value close to 1 means a strong fit. A value close to 0 means weak explanatory power. R^2 alone does not confirm that the model is correctly specified.

3.4 Adjusted R^2

Because R^2 increases whenever you add variables to a model, we also report the adjusted coefficient of determination.

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1},$$

where n is the number of observations and p is the number of predictors. A large gap between R^2 and R^2_{adj} suggests that extra variables are not adding meaningful explanatory power.

3.5 Root Mean Squared Error

Root mean squared error (RMSE) measures the average size of prediction error.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

Lower RMSE means better predictive accuracy. Because RMSE uses the same units as the response variable, it is easier to interpret practically than R^2.

3.6 Residuals

Residuals measure the difference between observed and predicted values.

$$e_i = y_i - \hat{y}_i.$$

If residuals scatter randomly around zero with no patterns, the linear model is a reasonable fit. Patterns suggest the model is missing important structure.

3.7 K Fold Cross Validation

To evaluate model stability, we used K fold cross validation with $K = 5$. We split the dataset into disjoint subsets $D_1, \ldots, D_K$. For each fold k, we trained the model on $D_{-k} = D \setminus D_k$ and evaluated it on D_k. The cross-validated R^2 is

$$R^2_{CV} = \frac{1}{K} \sum_{k=1}^{K} R^2_k.$$

This reduces dependence on any single train test split and gives a more reliable estimate of out of sample performance.

3.8 Random Forest

A random forest is a collection of decision trees $\{T_b\}_{b=1}^{B}$, each trained on a bootstrap sample of the data. For regression, the model averages predictions across trees.

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x).$$

Random forests do not assume a linear relationship between predictors and the response. That makes them useful as a comparison model. Feature importance values measure each variable's relative contribution to predictive accuracy, giving a different angle than linear regression coefficients.

Chapter 4

Analysis

4.1 Overview of Approach

We model obesity prevalence as a function of both socioeconomic variables and food access measures. That lets us compare their relative importance directly. We move through correlation analysis, linear regression, residual and cross validation evaluation, and a random forest comparison.

4.2 Correlation Analysis

Pairwise correlations show a clear split between socioeconomic and food access predictors. Percent without a high school diploma has the strongest positive correlation with obesity ($r \approx 0.75$), followed by poverty rate and percent uninsured. Food access variables like grocery density and distance to the nearest store show weaker and less consistent relationships. This suggests socioeconomic conditions associate more closely with health outcomes than physical food proximity does.

Figure 4.1 shows the full correlation matrix.

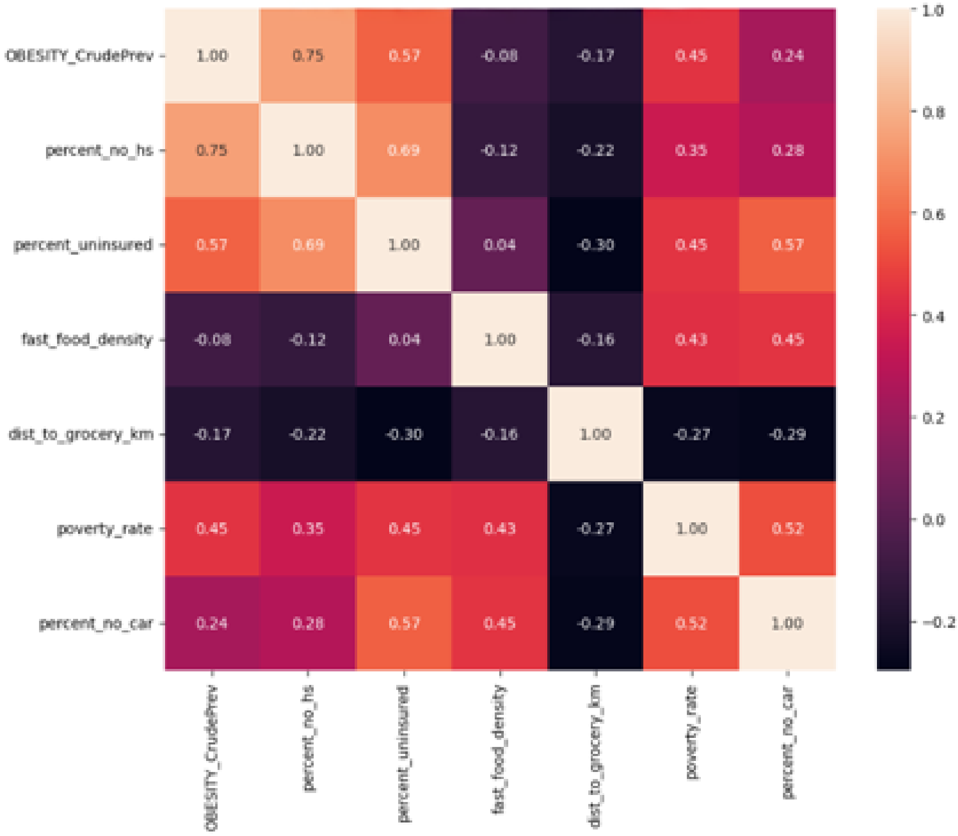

Figure 4.1: Correlation matrix of socioeconomic and food access variables

Figure 4.2 shows the strong positive relationship between lack of educational attainment and obesity prevalence.

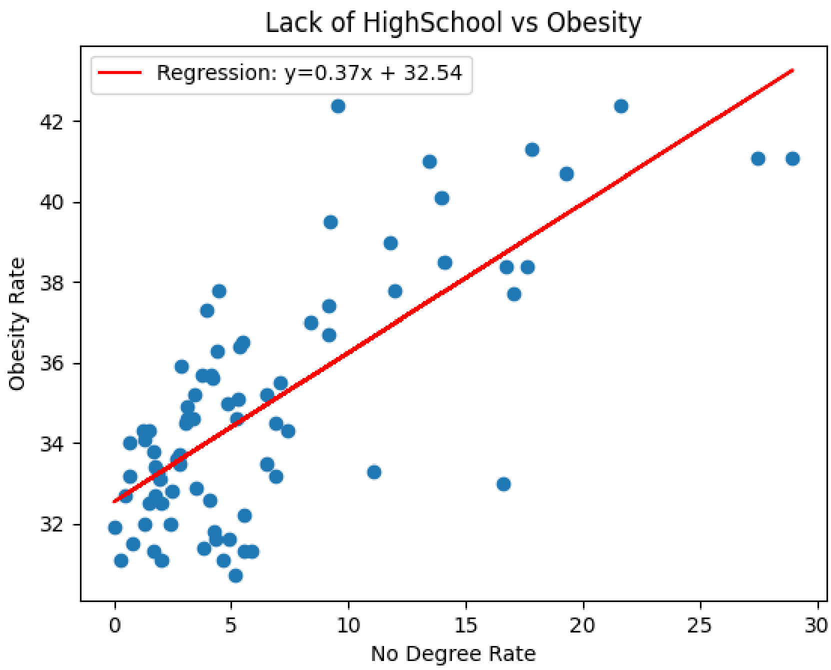

Figure 4.2: Scatter plot with regression line showing percent without a high school diploma versus obesity prevalence

4.2.1 Spatial Distribution of Obesity

Mapping obesity prevalence across Lancaster County census tracts shows that health outcomes are not randomly distributed. The highest vulnerability neighborhoods in the urban core align closely with the highest poverty concentrations and lowest educational attainment. The county-wide average obesity rate is 34.9%. Specific urban tracts consistently exceed 40%.

4.2.2 The Geometry Versus Functionality Problem

We initially treated straight-line distance from a census tract centroid to the nearest grocery store as the primary barrier to food access. Spatial analysis and regression testing showed this metric is a poor predictor of health outcomes. It yielded an $R^2 \approx 0.04$ and a near-zero correlation of $r = -0.19$.

That finding pushed us to rethink the approach. High obesity tracts appeared across all distance ranges, including tracts right next to grocery stores. Geometric proximity does not equal functional access. A 0.8-mile grocery trip is minor for someone with a vehicle. For a resident without a car, that same trip may require a 3.2-mile transit route with multiple bus transfers and about 45 minutes of travel time.

4.2.3 Transit Access and Socioeconomic Factors

To look at functional access more directly, we overlaid StarTran bus route data onto vulnerability heatmaps. We focused on high frequency corridors.

- **Route 44 (O Street).** This high coverage corridor cuts through Lincoln's most vulnerable tracts. Poor health outcomes persist there despite strong bus coverage. Transit availability alone cannot close the health gap.
- **Route 27 (North 27th St).** This route connects lower resource neighborhoods to affordable food outlets like Aldi. It provides a vital service but does not statistically reduce the high obesity rates in those tracts.

These findings confirm that barriers to healthy outcomes in Lancaster County are not purely spatial or transit-based. Financial and educational constraints shape them heavily. To capture this compounding effect, we built a **Neighborhood Vulnerability Index (V)**.

$$V = \frac{(\text{Fast Food Density}) \times (\text{Housing Cost Burden \%})}{\text{Distance to Healthy Grocery Store} \times \text{Transit Frequency}}$$

4.2.4 Transit Case Studies

Figure 4.3 shows how major transit lines intersect with areas of high poverty concentration in Lincoln's urban core.

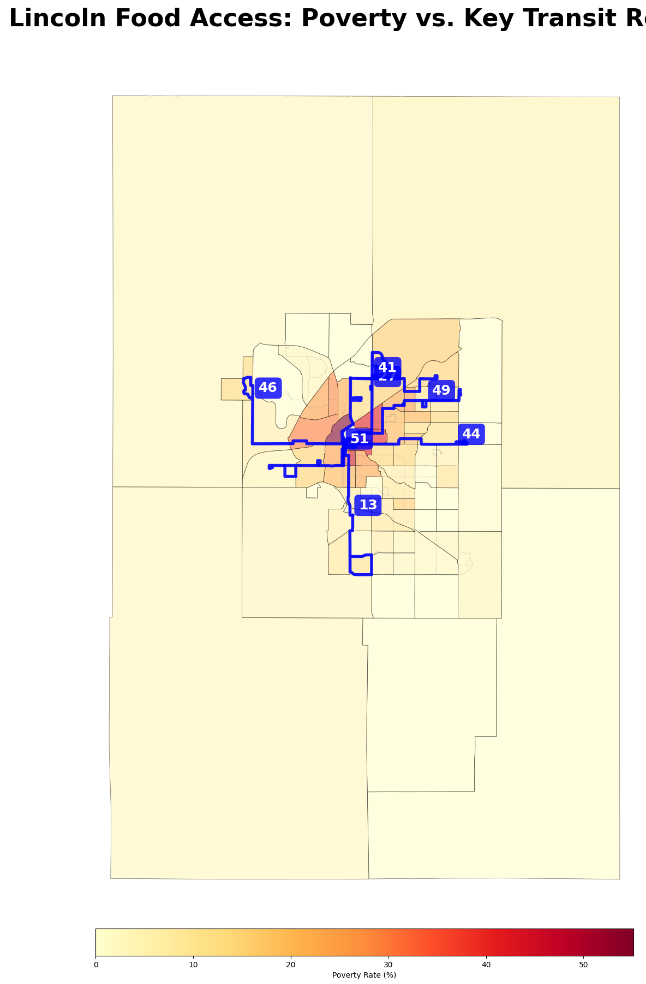

Figure 4.3: Lincoln food access overview showing poverty rate by census tract with key StarTran transit routes

Figures 4.4 and 4.5 show two transit corridors that serve as critical connections for residents in high vulnerability tracts.

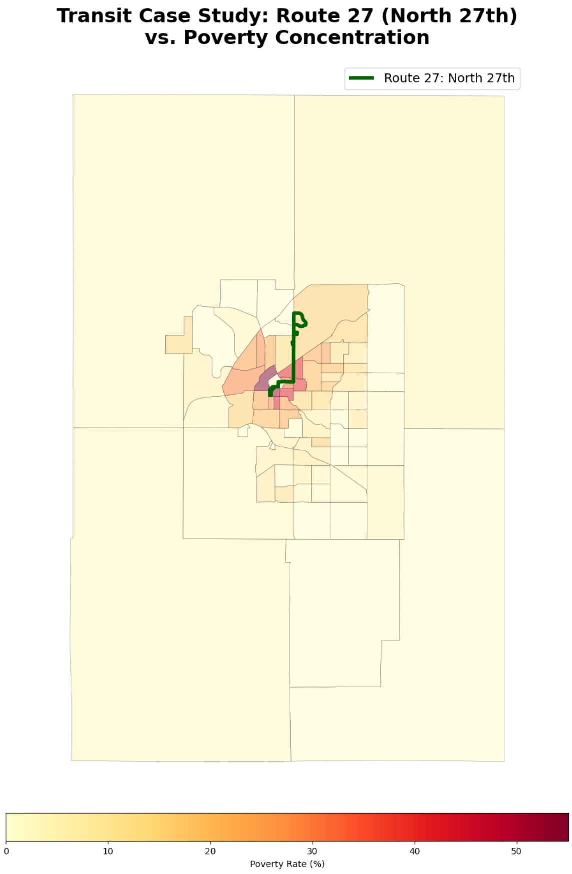

Figure 4.4: Route 27 (North 27th St) case study

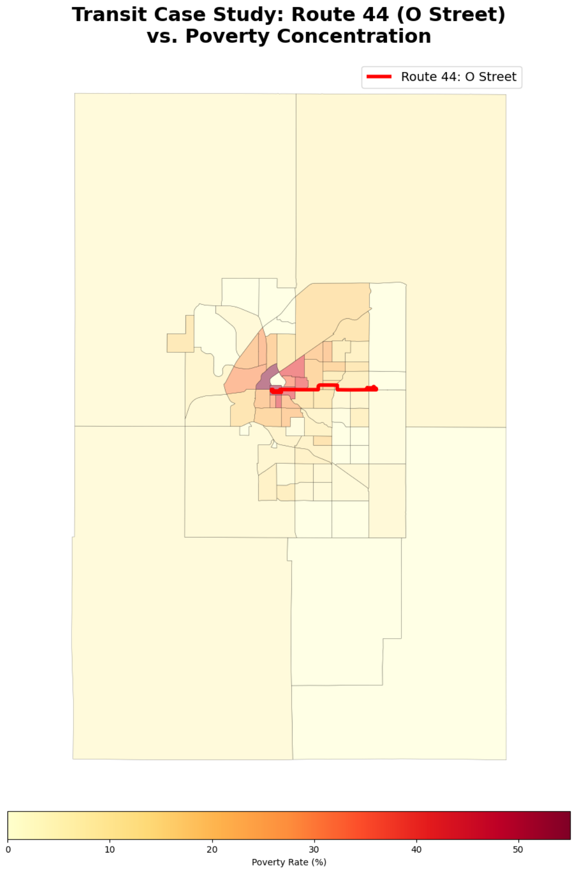

Figure 4.5: Route 44 (O Street) case study

Routes like North 27th and O Street connect high poverty neighborhoods to grocery outlets. But residents who depend on fixed transit schedules face real friction, especially during off-peak hours or when carrying groceries.

4.2.5 Statistical Validation of Spatial Patterns

Our statistical models support the spatial clusters visible in the maps. A multiple linear regression model applied to these tracts achieved $R^2 \approx 0.49$. Socioeconomic constraints explain nearly half of the variation in obesity. Educational attainment emerged as the dominant predictor. Percent without a high school diploma showed a correlation of $r \approx 0.75$ with obesity.

A random forest model raised explanatory power to $R^2 \approx 0.59$. Feature importance analysis confirmed education as the top predictor. Lack of a high school diploma carried an importance weight of 0.721, far above physical distance to a grocery store at 0.143. That consistency across both linear and nonlinear models strengthens the overall conclusion.

Figure 4.6 shows the spatial distribution of obesity across Lancaster County census tracts.

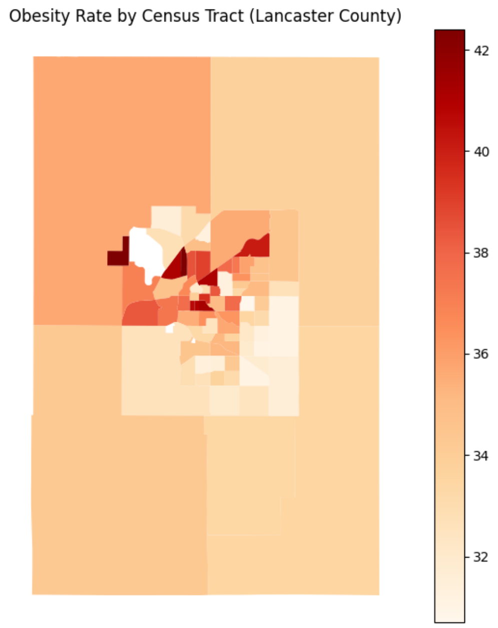

Figure 4.6: Obesity prevalence by census tract in Lancaster County

4.3 Linear Regression Model

We fit a multiple linear regression model of the form

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i,$$

where the predictors represent percent without a high school diploma, distance to the nearest grocery store, and fast food density. The fitted model achieves $R^2 \approx 0.49$ and RMSE ≈ 0.66. Since obesity is measured as a percentage, the RMSE means predicted values are typically within less than one percentage point of observed values. That is a relatively close fit on this data split.

Figure 4.7 compares predicted and actual obesity values. The model captures the general trend, but the spread of points and presence of outliers show that not all variation is explained.

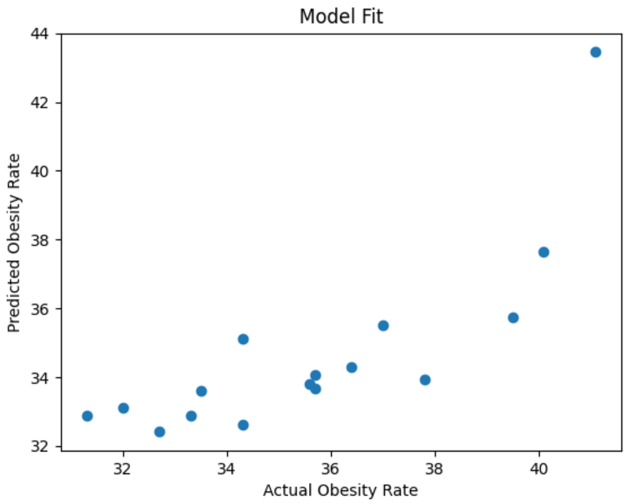

Figure 4.7: Predicted versus actual obesity values from the linear regression model

4.3.1 Model Considerations and Evaluation

R^2 measures the proportion of variance the model explains, but it can mislead you if the functional form is wrong. We also report adjusted R^2 to account for model complexity. Correlation analysis showed moderate associations among some socioeconomic variables, which signals potential multicollinearity. We treat individual coefficients with caution and put more weight on overall model behavior and relative predictor importance.

We considered alternative specifications. Because obesity prevalence is a proportion, fractional logistic regression could be appropriate in future work. A log transformation of distance might better capture diminishing access effects. We kept the linear model here for interpretability and because it gives a reasonable approximation of the observed data.

4.4 Residual Analysis

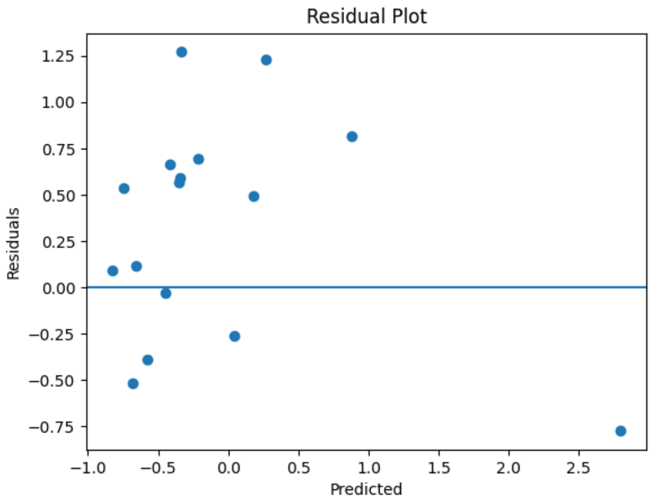

Figure 4.8: Residual plot for the linear regression model

The residuals in Figure 4.8 scatter mostly randomly around zero. That suggests the linear model is a reasonable approximation. One notable outlier indicates the model does not fully capture variation in all tracts.

4.5 Cross Validation

Repeated 5 fold cross validation yields a mean $R^2 \approx 0.33$ with meaningful variance across folds. This is lower than the single split result, which makes sense given the small number of census tracts and the complexity of health outcomes. The result shows the model identifies real patterns in the data but with limited stability. Adding variables like age structure, affordability measures, or more detailed transportation data would likely improve predictive stability.

4.6 Random Forest Comparison

A random forest model using 100 trees achieves $R^2 \approx 0.59$, confirming that nonlinear structure exists. The relative importance of predictors stays consistent with the linear model. Table 4.1 shows that percent without a high school diploma dominates in both models, while food access variables contribute little.

Feature	**Linear Regression Coefficient**	**Random Forest Importance**
Percent No High School	0.784	0.721
Distance to Grocery (km)	0.024	0.143
Fast Food Density	0.014	0.136

Table 4.1: Comparison of predictor influence across linear regression and random forest models

Regression coefficients and random forest importance values are not directly comparable because they measure different aspects of predictor influence. Both models identify educational attainment as the dominant factor. We also tested poverty rate and percent uninsured as alternative socioeconomic predictors. Exact rankings shifted slightly but the broad pattern held. Socioeconomic variables consistently outperformed food access measures across all specifications.

Chapter 5

Results and Conclusions

5.1 Correlation Patterns

The correlation matrix shows a clear split between socioeconomic and food access predictors. Percent without a high school diploma has the strongest positive relationship with obesity ($r \approx 0.75$). Poverty rate and percent uninsured also show moderate positive correlations. Distance to grocery stores and fast food density show weaker and less consistent relationships. Physical food access does not strongly explain variation in health outcomes at the census tract level.

5.2 Regression Findings

The multiple linear regression model explains roughly half the variation in obesity rates ($R^2 \approx 0.49$) with an RMSE of about 0.66 percentage points. Percent without a high school diploma carries a substantially larger coefficient than food access variables. Education level is the stronger predictor.

5.3 Model Stability

Cross validation results show a mean $R^2 \approx 0.33$ with meaningful variance across folds. The model captures real relationships but is sensitive to data partitioning. Residual analysis supports the use of the linear form as a reasonable approximation, with one notable outlier.

5.4 Comparison of Models

The random forest achieves higher explanatory power ($R^2 \approx 0.59$) but reaches the same substantive conclusion. Educational attainment accounts for most of the predictive importance (0.721). Physical distance to grocery stores contributes far less (0.143). Agreement across modeling approaches strengthens confidence in the finding.

5.5 Key Takeaways and Implications

Socioeconomic conditions, particularly educational attainment, predict chronic health outcomes in Lancaster County better than physical proximity to food. Even in areas with reasonable geographic food access, underlying economic and educational barriers limit residents' ability to benefit from nearby resources.

Public health policy needs to reflect this. Efforts focused only on placing new grocery stores are unlikely to improve chronic health outcomes unless they also address income, education, and transportation access. These are the structural conditions that shape whether residents can actually use those resources.

5.5.1 Spatial Context: Transportation and Food Access

Figure 5.1 overlays bus routes on a heatmap of households without vehicle access. This gives spatial context for food accessibility beyond simple distance.

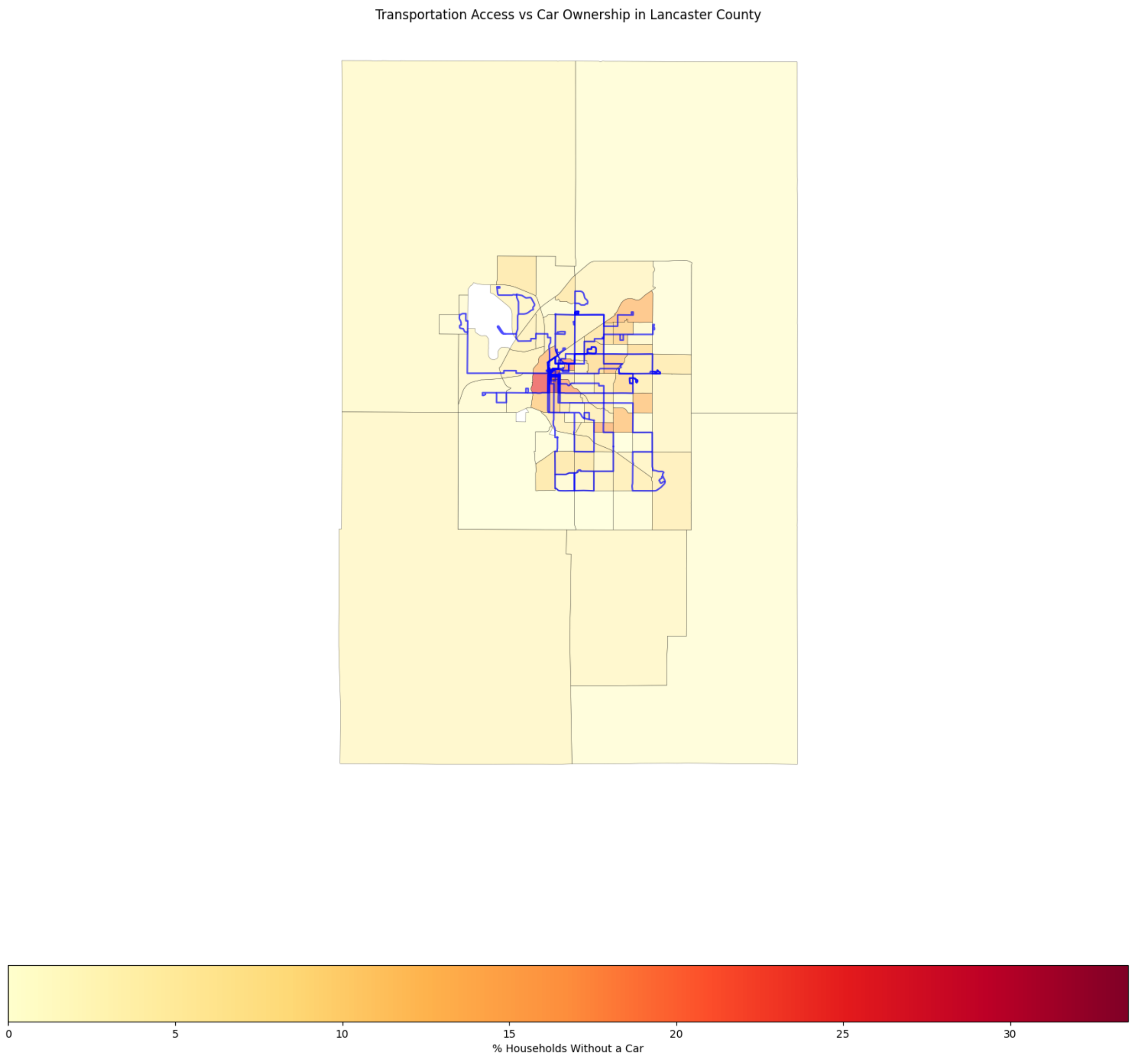

Figure 5.1: Bus routes overlaid on percent of households without a car, by census tract

Many low car ownership areas have public transit routes running through them, especially in the densely populated urban core. Some residents can physically reach grocery stores without a vehicle. But poor health outcomes persist in those same areas. Transit availability alone does not remove barriers to healthy food access. Socioeconomic factors play the dominant role.

5.6 Future Directions

Prior research on obesity in Nebraska suggests age-related and child-focused variables could improve the model's explanatory power [11]. Variables like SNAP participation, free and reduced lunch eligibility, school meal program access, and the percentage of children in single parent households could give a clearer picture of how socioeconomic conditions affect different populations. Future work could add these variables alongside demographic factors like age distribution. That would enable more targeted public health strategies for both urban and rural communities.

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